Murat Akman

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Research interests

Harmonic analysis, partial differential equations, geometric measure theory, potential theory and convex geometry. My work focuses on boundary measures, PDEs in rough domains, and Brunn–Minkowski inequalities and Minkowski problems for nonlinear capacity.

Nineteen published works (eighteen journal articles and one research monograph) and one submitted preprint. EPSRC New Investigator Award holder; corresponding organiser of an Isaac Newton Institute programme in 2027; Chair of the British Mathematical Colloquium 2028.

Education and appointments

- **Senior Lecturer in Mathematics**, University of Essex, 2022–present.
- **Lecturer in Mathematics**, University of Essex, 2019–2022.
- **Evarist Giné Assistant Research Professor**, University of Connecticut, 2016–2019. A named three-year research professorship, filled by open international competition.
- **Postdoctoral Researcher**, ICMAT–CSIC, Madrid, 2014–2016.
- **Research programme member**, MSRI/SLMath, Berkeley, 2017; Institut Mittag-Leffler, 2013.
- **PhD in Mathematics**, University of Kentucky, 2014. Adviser: John L. Lewis.
- **BS 2006 and MS 2009 in Mathematics**, Middle East Technical University, Ankara.

Selected research contributions

- **Elliptic PDEs in rough domains.** With Hofmann, Martell and Toro: perturbation and boundary estimates on 1-sided NTA domains satisfying the capacity density condition, without assuming Ahlfors-regular surface measure. *Forum Math.* and *Adv. Calc. Var.* (2023).
- **Rectifiability and harmonic measure.** With Badger, Hofmann and Martell: a characterisation of rectifiability through absolute continuity of surface measure with respect to harmonic measure on 1-sided NTA domains with Ahlfors–David regular boundaries. With Bortz, Hofmann and Martell: interior approximation results. *Trans. AMS* (2017); *Ark. Mat.* (2019).
- **Local boundary geometry.** With Azzam and Mourgoglou: absolute continuity of harmonic measure on boundary pieces under local content-thickness and Ahlfors-regular separating-set hypotheses. *Adv. Math.* (2019).
- **Nonlinear boundary measures.** With Lewis and Vogel: dimension and σ-finiteness results, and counterexamples to Fatou-type theorems. Sole-authored work on planar measures and failure of absolute continuity in Reifenberg-flat domains. *Adv. Math.* (2017); *J. Anal. Math.* and *Comm. PDE* (2022); *Ann. Acad. Sci. Fenn.* (2014); *Pacific J. Math.* (2017).
- **Nonlinear capacity and convex geometry.** With Gong, Hineman, Lewis and Vogel: a Brunn–Minkowski inequality, its equality case and the associated Minkowski problem (*Memoirs AMS*, 2022). With Mukherjee: existence for the Minkowski problem for p-harmonic measure (*Calc. Var. PDE*, 2025). With Aguas-Barreno and Mukherjee: a local Brunn–Minkowski inequality for $2 < p < n + 1$ (submitted, 2026).

Grants and funding

Held as Principal Investigator

- **EPSRC New Investigator Award**, £311,464 full economic cost, 2022–2025: *Stability of Brunn–Minkowski inequalities and Minkowski type problems for nonlinear capacity*. Funded a three-year postdoctoral appointment, which I recruited and line-managed, and produced the 2025 *Calc. Var. PDE* paper.
- **London Mathematical Society Scheme 1**, £5,430, for *PDEs@Essex 2026*.
- **Heilbronn Institute Small Grant**, £4,880, for *PDEs@Essex 2026*.

Under review

- **Royal Society Faraday Discovery Fellowship**, Principal Investigator, £3.5M over ten years, host University of Essex. Stage 1 application under review.
- **Royal Society Newton International Fellowship 2026**, £255,326, to host Dr Marcin Gryszówka at Essex. I am the UK sponsor and host.
- **Turkish Research Council Postdoctoral Fellowship 2026**, £190,112 sterling equivalent, to host Dr Idem Susuzlu of Izmir Institute of Technology at Essex.
- **EPSRC Small Grant**, Principal Investigator, £72,723.

Fellowship shortlisting

UKRI Future Leaders Fellowship, Round 9 — selected for final interview (£1.45M proposal).

Teaching, supervision and researcher development

Teaching

- Teach 2.5 to 3 full modules a year at Essex, in analysis, differential equations and mathematical physics.
- Student feedback for Complex Variables (MSc): 4.8/5 in 2024–25 and 4.6/5 in 2025–26 for organisation, explanation and approachability; my Level 5 core module in 2025–26 scored above the School's Level 5 average on every item.
- **Senior Examinations Officer**, School of Mathematics, Statistics and Actuarial Science, 2020–2022.
- **Deputy Director of Postgraduate Research**, 2022–2023.
- I founded my School's postgraduate research seminar.

Research supervision

- **Ariel A. Aguas-Barreno**, doctoral student, since October 2023. Co-author of the 2026 submitted preprint. He has given seminar and conference talks and is in his completion period (he will defend in March 2027).
- **Dr Shirsho Mukherjee**, postdoctoral researcher, 2022–2025, recruited on my EPSRC award. Co-author of the 2025 *Calc. Var. PDE* paper. Now at CUNEF Universidad, Madrid. His own EPSRC fellowship proposal reached interview.
- **External PhD examiner**: Universidad Autónoma de Madrid; University of Birmingham.

Widening participation and early-career development

- **Nuffield Research Placements**: two school students each summer since 2023, eight in total. The scheme prioritises students from low-income households and families with no history of university attendance.
- Supported four fellowship applications by early-career researchers. One candidate was awarded a Royal Society Newton International Fellowship in 2022 and took a permanent post elsewhere.
- I deliver internal Grant Writing Days across two Schools at Essex, and I host annual EPSRC Portfolio Manager visits that bring funders directly to researchers.
- Co-led the application to host a week-long **Academy for PhD Training in Statistics** at Essex from 2027, for about 100 doctoral students from across the UK. This will be hosted at Essex in July 2027.

Invited lectures

Plenary lectures: LMS Harmonic Analysis and PDE Research Network, Kent (2021); *Nonlinear PDE, Calculus of Variations and Applications*, Sussex; Northeastern Analysis Meeting, Albany (2017); *Nonsmooth Analysis*, Connecticut (2017); Workshop on Harmonic Analysis, PDEs and GMT, ICMAT Madrid (2015). Further invited lectures at the Hausdorff Research Institute for Mathematics, Bonn, and at FBP2020.

Invited talks: more than 50 since 2014, including colloquia and seminars in the UK, Europe and the USA at Oxford, Warwick, Bristol, Bath, Edinburgh, Heriot-Watt, Bonn, Uppsala, IMPAN (Poland), Brown, Minnesota, Pennsylvania, Tennessee and SLMath Berkeley.

Research leadership and community organisation

- **Corresponding organiser**, Isaac Newton Institute programme *Frontiers in Harmonic Analysis, Elliptic and Parabolic PDEs, and GMT*, Cambridge, January–April 2027. I coordinate the scientific programme and the invitations for this competitively selected four-month programme. About 150 participants are expected.
- **Chair, British Mathematical Colloquium 2028**. This will be hosted at Essex in July 2028.

- **Founder and organiser, *PDEs@Essex***, an international workshop series held in 2022 and every year since 2024. Speakers have included Sir John Ball FRS, former President of the London Mathematical Society and of the International Mathematical Union. The workshop reserves places for early-career researchers and is funded by the LMS, the Heilbronn Institute and EPSRC.
- **Special sessions organised** at AMS Sectional Meetings (*Regularity Theory of PDEs*; *Regularity of PDEs on Rough Domains*); a graduate student conference at Connecticut, 2017–2019.
- Member of the Eastern Arc EPSN group (Sussex, Kent, UEA, Essex), which develops collaborative bids.

Peer review and professional service

- Member, **UKRI Future Leaders Fellowship Peer Review College**, 2025–present, and **EPSRC Peer Review College**, 2020–present.
- **EPSRC panel member**: Postdoctoral Fellowships; Open and Open Plus Fellowships; the EDI Hub+ Flexible Fund. Member of the EPSRC Fellowships Focus Groups.
- Referee for *Inventiones Mathematicae*, *Memoirs* and *Transactions of the AMS*, *IMRN*, *Israel J. Math.* and other journals.
- **London Mathematical Society Representative** for Essex, 2026–present.
- Essex nominee to the **EPSRC Roundtable on University Selection Processes** (inclusive selection).

Institutional roles

- **Director of Research**, School of Mathematics, Statistics and Actuarial Science, 2023–present. Research strategy, grant development and mentoring for about 50 academics and their doctoral community.
- **REF2029 UoA10 Lead**, 2023–present, responsible for Essex's UoA10 submission.
- Member, **University Academic Staffing Committee** (promotion and permanency across the University), 2023–present.
- **Deputy Head of School**, 2023–2025. Managed the School's research expenditure budget of about £250,000 a year and its allocation decisions.
- Equality, diversity and inclusion: my action plans contributed to the School's **Athena SWAN Silver Award**. Public engagement: open and applicant days; Nuffield placements.